

MINERVA CryoCell P&ID — Interactive Viewer Manual (v4)

Project: MINERVA CryoCell — SCK CEN (MYRRHA/MINERVA Phase 1)

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Standard: SCK CEN AD_01.16 · Status: S2 — FOR ACCEPTANCE · RESTRICTED

1. What it is

`HTML_INTERACTIVE/` is a self-contained, browser-based viewer for the four MINERVA CryoCell P&ID sheets. It lets a reviewer explore the layered SVGs without any CAD software: toggle layers, apply default views, search tags, switch colour/mono, zoom, pan and export PNGs.

HTML_INTERACTIVE/		
	<code>index.html</code>	 landing page (sheet gallery)
	<code><sheet>.html (x4)</code>	 per-sheet interactive viewer
	<code>css/viewer.css</code>	
	<code>js/pid-viewer.js</code>	 layer control, views, zoom/pan, export
	<code>js/search.js</code>	 tag search & highlight
	<code>assets/</code>	
	<code>*_v4.svg (x16)</code>	 the drawing layers
	<code>thumbs/*.png (x4)</code>	 gallery thumbnails

2. Launching the viewer

Because the viewer loads SVG assets via `fetch()`, it must be served over HTTP — opening `index.html` directly with a `file://` path will be blocked by the browser's same-origin policy.

From inside the `HTML_INTERACTIVE/` directory run a simple static server, e.g.:

```
cd HTML_INTERACTIVE
python3 -m http.server 8137
```

Then open `http://localhost:8137/` in your browser.

Note on localhost: When this viewer is launched on the Abacus AI Agent machine, that localhost refers to localhost of the Abacus AI Agent computer running the application, **not your local machine**. To run it yourself, download all files using the **Files** icon at the top right, navigate inside the downloaded folder, and serve the `HTML_INTERACTIVE` directory locally as shown above.

3. Landing page

The landing page (`index.html`) shows a gallery of the four sheets with thumbnails and metadata (drawing number, title, revision). Click any sheet to open its interactive view.

4. Controls (per-sheet view)

4.1 Layer panel

- Each of the 24 layers has a checkbox; class layers show a **colour swatch** matching the printed legend (see Colour & Legend Guide §2).
- Layers are grouped (frame/title, process pipes, equipment, instruments, annotation, reference) for quick navigation.
- Toggling a layer immediately shows/hides it in the drawing.

4.2 Default views (presets)

- The **Views** selector applies one of the five presets: FULL, PROCESS, CONTROL, MAIN, PRINT_MONO (documented in `DEFAULT_VIEWS_DOCUMENTATION.md`).
- Selecting a preset updates the layer checkboxes; you can then fine-tune individual layers on top of the preset.

4.3 Tag search

- Type a tag (e.g. `CV500`) in the **Search** box.
- Matching tags are highlighted and the view **pans/zooms to the first match**.
- Use the next/prev controls (or repeated Enter) to cycle through multiple matches.

4.4 Mono toggle

- Switches the drawing between **colour** and **monochrome** rendering. In mono mode, classes remain identifiable via inline line names and line-weight/dash differences.

4.5 Style toggle

- Switches between the filled/coloured presentation and a line-weight-only presentation.

4.6 Zoom & pan

- **Scroll wheel** (or zoom buttons) to zoom; **click-drag** to pan.
- A reset/fit control returns to the full-sheet view.

4.7 Valve overlay

- Enabling the `08B_Valves_HORIZONTAL_OVERLAY` layer shows the alternative horizontal valve representation on top of the standard symbols (off by default).

4.8 PNG export

- The **Export PNG** button rasterises the current view (with the current layer visibility, mono/style state and zoom) and downloads it as a PNG.
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5. Tips

- Start from a **default view**, then toggle individual layers for a focused review.
 - Use **PRINT_MONO** + **Export PNG** to verify how a sheet will look on a black-and-white A3 plot before issuing.
 - Tag search is the fastest way to locate a specific valve or instrument across a busy sheet.
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6. Relationship to the SVG/PDF set

The viewer renders the same `*_v4.svg` files delivered in `output_v4/`. The PDF set (A3 landscape, 420 × 297 mm) is the formal issue format; the HTML viewer is a review aid. Layer names, default views and class colours are identical across all formats.